Cross-Asset Signals for FX Futures Trading

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Strategy overview

- Use excess equity market returns to identify FX futures opportunities
- Generate signals by isolating FX movements related to country's equity performance
- Use strong signals to make trades
- Generates a positive return but relatively large risk stats and macro exposure make strategy undesirable

Motivation

- Strong foreign equity performance attracts capital
- Investing in foreign markets requires buying and transacting in the local currency
- Problems:
  - **Exchanging Currencies is time consuming and inefficient because of how FX markets work**
  - **FX rate appreciates when investors rebalance in the same direction and the market cannot absorb order flow**
- Goal: Profit off of the lead-lag relationship caused by such problems

Trading Hypothesis 1

- There exists a lead-lag relationship between a country's stock indices and the exchange rate of the country's currency
  - **Ex: A strong performance of the Nikkei 225 will be followed by an appreciation of the yen in FX markets**
- Economic Intuition: A strong performance in a country's equity markets will attract international investors looking to take advantage of a bull market. These investors will need to convert to the foreign currency, driving up the exchange rate

Trading Hypothesis 2

- The lag in the relationships previously mentioned, will be larger in less liquid FX markets
  - **Ex: The lag in the currency appreciation in Zimbabwe will be larger than the lag in Great Britain**
- Economic Intuition: Less liquid FX markets adjust more slowly to new information because trading depth is limited. As a result, capital flows take longer to move prices, leading to larger lags in exchange rate responses

Data

Data Sources

- CDS Data: Bloomberg
- Equity Data: Bloomberg
- FX Spot Rate Data: FRED
- FX Future Data: CME

Investment Universe

- FX Futures (CME)

Implementation Overview

Stage 1

- Our hypothesis relates FX rates to equities
- Need the movement in FX rates attributable to equities to train our models
- Get this movement by hedging out factors unrelated to equity

- Stage 2:
- Train models to predict this movement
  - Models include:
    - OLS
    - Random Forest

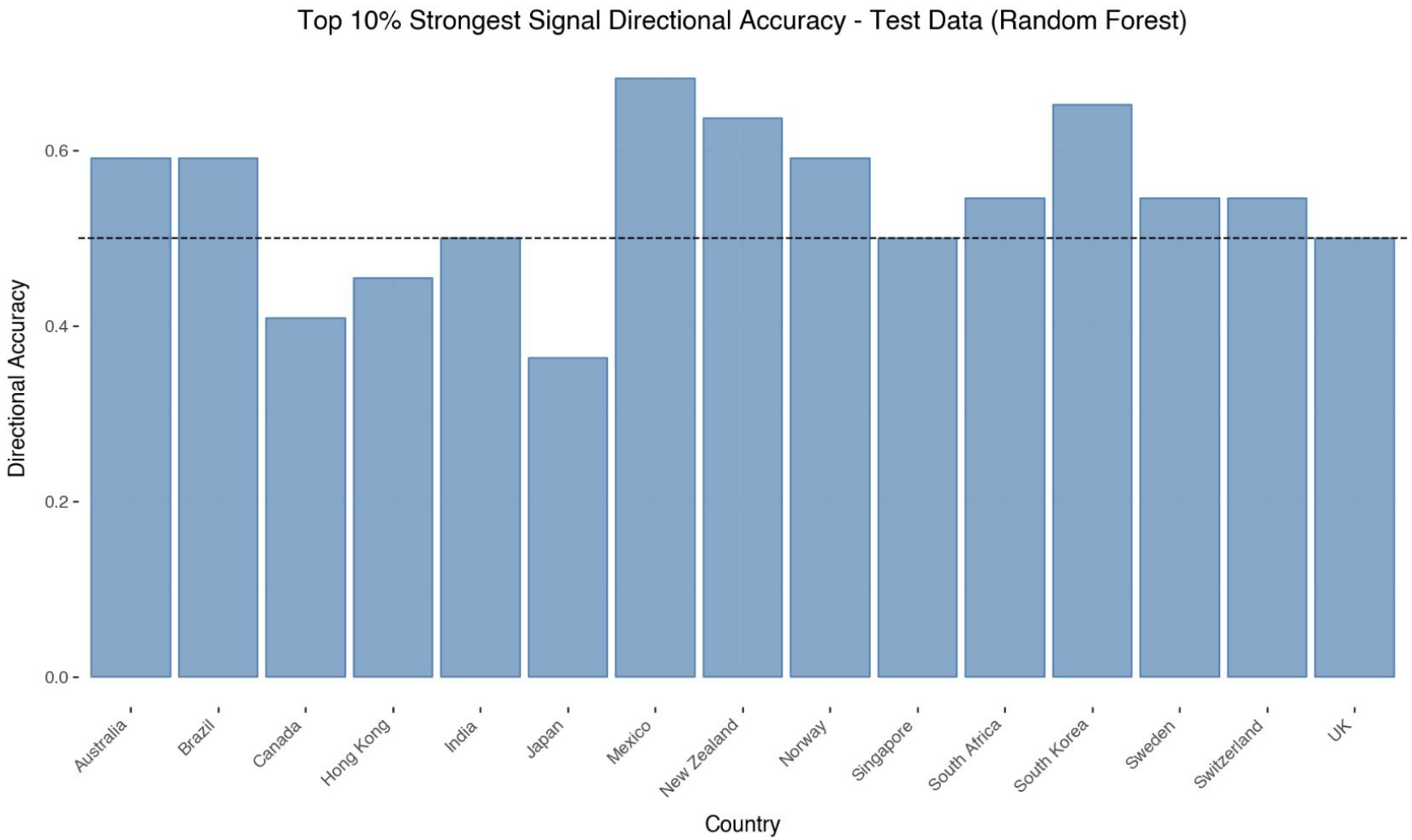
Stage 1 - Hedging

- FX carry: daily forward implied yield for each currency pair
  - **Statistically significant and hedged out**
- The Dollar factor: movements driven by dollar strength or weakness
  - **Statistically significant**
  - **Each currency has a specific dollar hedge ratio**
- CDS: controls for country-specific credit risk and macro distress
  - **not statistically significant**
  - **implies perceived credit risk has little effect on FX rate**
- Left with FX movement attributed to equity performance

Stage 2 - Prediction Models

- OLS Model:
  - **No linear relationship between isolated FX movement and equities**
  - **directional accuracy for the 10% strongest signals was less than 50%**
- Random Forest Model
  - **Trained on:**
    - Excess Returns
    - rolling average of returns and volatility
- Directional accuracy for strongest predictions was 55%

Stage 2 - RF Model (cont)

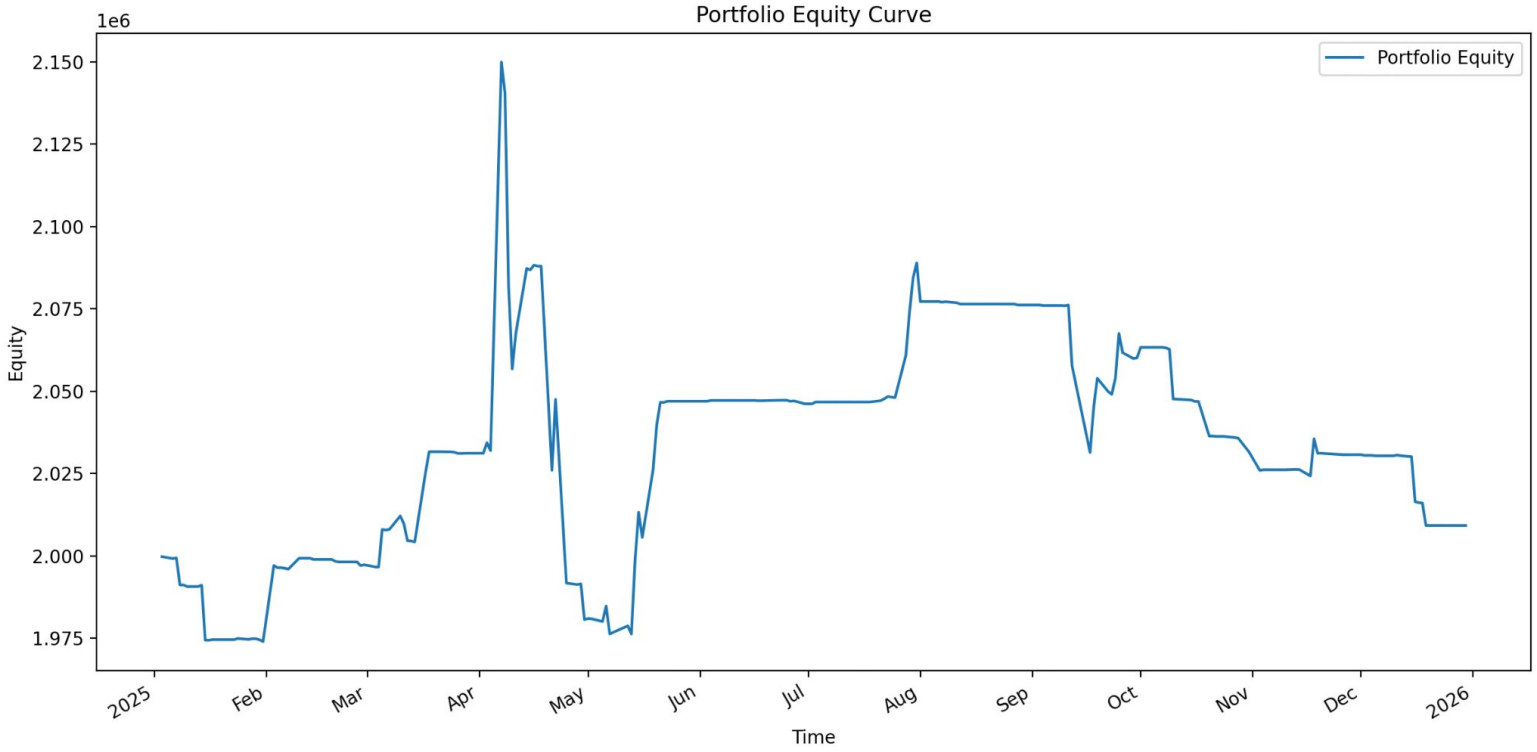


Backtesting

- Entry and Exit thresholds given by signal strength
- Hedge exposure to US dollar via ETF
- Leverage taken at a 5-1 ration on $2 million USD
- Transaction costs of $0.85/contract are enforced for each position change
- Strict timing enforced to ensure no look ahead bias

Strategy PnL - statistics

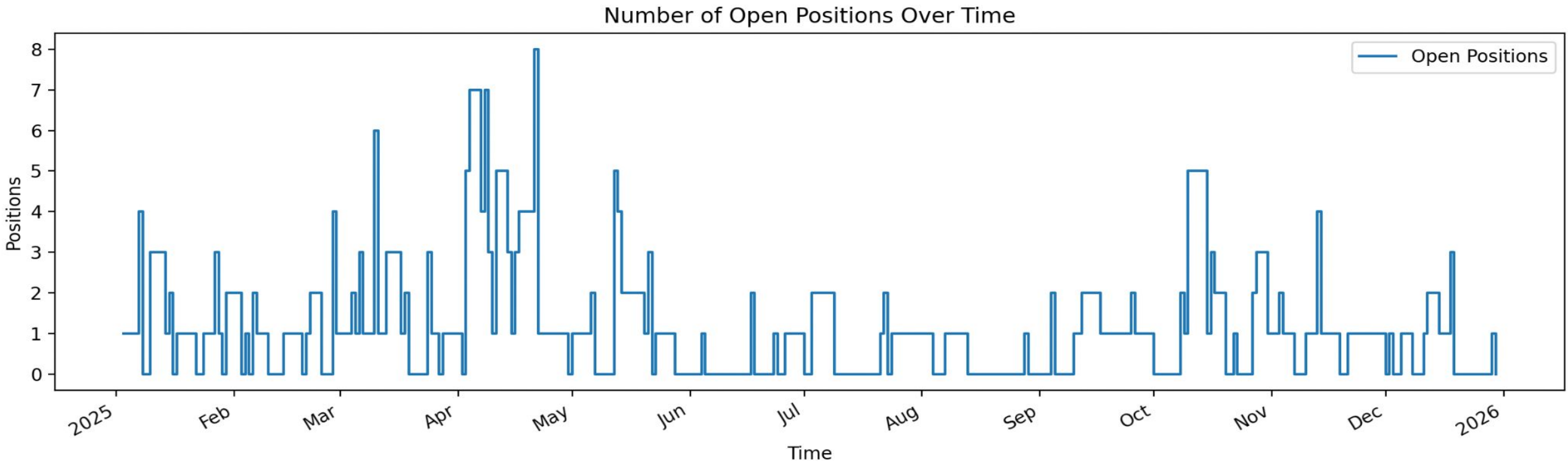
- Total return: .47%
- Annual Vol: 9.47%
- Sharpe Ratio: 0.1048
- Win Rate: 39.86%
- Max Drawdown: -8.08%



PnL Analysis

- Returns (0.47%) are mediocre, but notably positive
- Strategy is very volatile (9.47%) compared to returns
  - **Specifically volatile during april**
  - **likely because of macro factors (liberation day tariffs)**
- The Max drawdown (-8.08%) is not too large in general but is large compared to returns

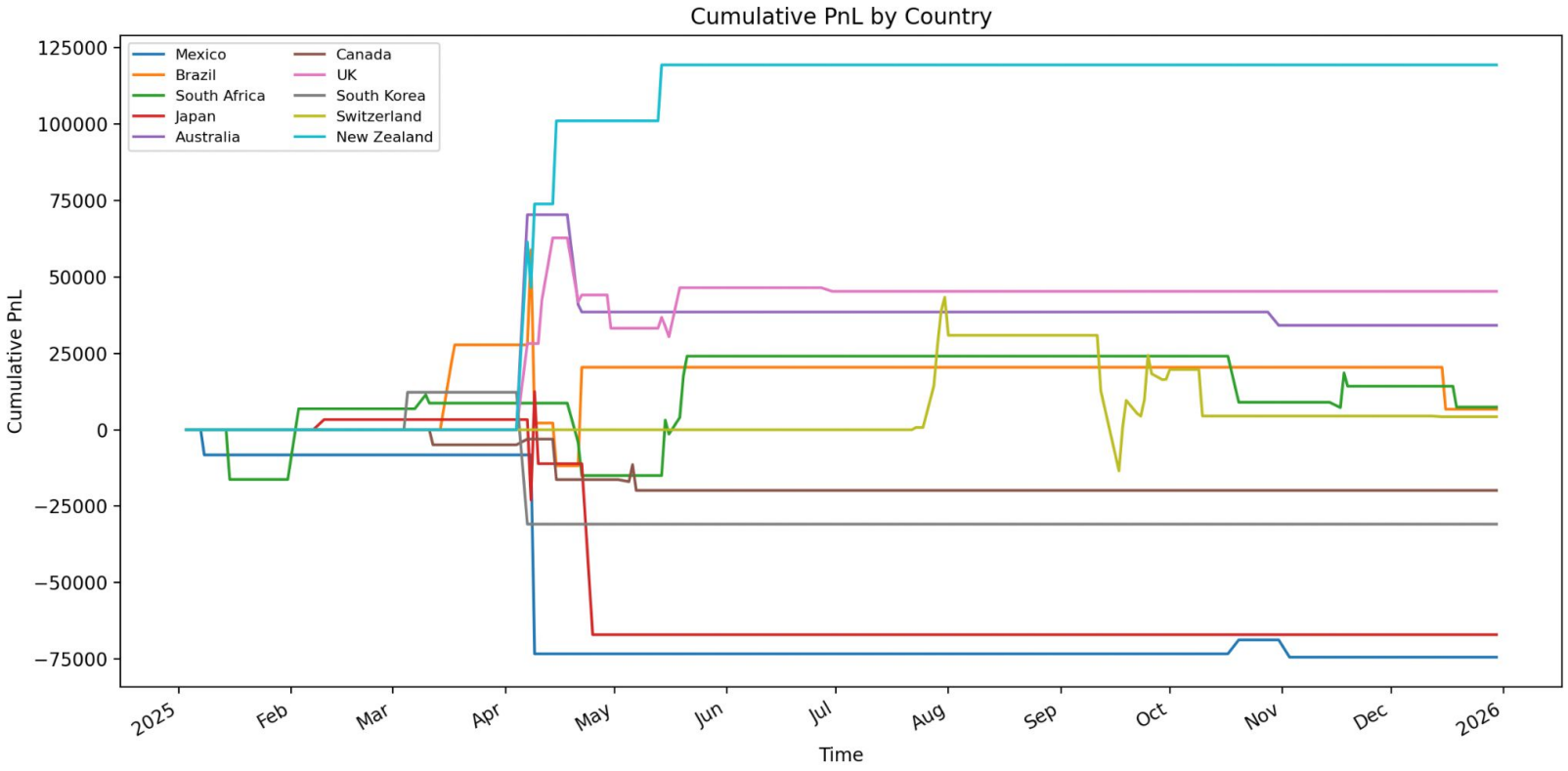
Position Analysis



- At 217 total trade, we make slightly less that one trade per day
- Our trades have a win rate of 40%, but have an average win of 40% and loss of 25%

Hypothesis 2

- Second hypothesis related lag length to liquidity
  - **implies we should make more money in less liquid markets**
- Reality: No relationship between liquidity and results



Risk Analysis

- Implemented a Stop loss when Margin Called
  - **Not effective during April because we are already up**
- Volatility (9.47%) and Max drawdown (-8.08%) are high relative to returns
- Attempted to hedge against global factors but these hedges proved ineffective
  - **During April the macro factors greatly affected our strategy**

Strategy Summary

- Returns are positive, but mediocre
- Relatively high volatility and max drawdown
- Consistent trading with a lower win percentage
- Attempted to hedge out macro factors, but failed
- Recommendation: Don’t Invest

References

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